# Furukawa Minda Electric (FME)

## Man-Machine Interlink (MMI) Checksheet & Operations Workflow Guide

**Production Architecture**
React 19 • Vite • Tailwind CSS

**System Purpose:** Replaces manual paper shop-floor clipboards with an interactive, digital factory check-sheet engine. It covers 31 automated Cutting & Crimping machines, Resistance Welding, and Manual Crimping units at the Bawal manufacturing plant (with expansion support for the Gujrat facility).

**Scope of this Document:** Outlines the end-to-end operational flow of the system and provides a clean inventory explaining why every file exists and what function it performs.

## 1. How The Project Starts & Execution Lifecycle

```
Browser loads index.html
   ↓
Loads /src/main.jsx (React createRoot mounts to #root)
   ↓
App.jsx sets up <AuthProvider> (loads active operator session from sessionStorage)
   ↓
BrowserRouter matches URL route (default / redirects to /dashboard)
   ↓
MainLayout.jsx renders:
  • Left Sidebar (collapsed/expanded)
  • Top Navbar (breadcrumbs + operator badge)
  • Routed page <Outlet /> (or switches to 100% full-screen if on a checksheet)
```

- **Step 1 (Mounting):** `index.html` loads Google Fonts and launches `src/main.jsx`.
- **Step 2 (Auth State):** `src/context/AuthContext.jsx` checks `sessionStorage` for `fme_operator_session`. If credentials exist, the user stays authenticated across route switches.
- **Step 3 (Route Resolution):** `src/App.jsx` listens to URL changes and routes requests to the Dashboard or Checksheet selectors.
- **Step 4 (Smart Layout Adjustment):** `src/MainLayout.jsx` inspects the route depth. When a user is filling an actual checksheet (5+ URL segments), it automatically hides the Sidebar and Navbar to maximize table viewing width on shop-floor tablets.

## 2. Complete Project Workflows

### Workflow A: Operations Dashboard & Machine Telemetry

Plant managers analyze daily output throughput and runtime hours across customizable multi-day windows.

```
User visits /dashboard
   ↓
OperationsDashboard.jsx renders top cascading filter bar (Unit → Section → Process → Machine → Days)
   ↓
Component calculates deterministic period telemetry (HOP units/hr, daily runtime hours, uptime ratio)
   ↓
Renders 3 dynamic SVG bar charts:
  1. Combined Dual-Bar Graph (Hourly Output vs Machine Runtime per day)
  2. Dedicated HOP Analysis Chart (Daily output rate & peak performance)
  3. Machine Runtime Hours Trend (Active hours out of 24h & uptime efficiency)
```

### Workflow B: Plant → Department → Machine Selection

Operators drill down through the factory hierarchy to access their specific machine.

```
User clicks 'Machine Checksheet' (/machine-checksheet)
   ↓
UnitSelectionPage.jsx: User selects plant ("Bawal Unit" or "Gujrat Unit")
```

```
   ↓
OperationalUnitSelectionPage.jsx: User selects department (CNC Machining, Assembly, or SRC Quality)
   ↓
OperationalUnitRouter.jsx: Routes to CnCLayout.jsx, AssemblyLayout.jsx, or SrcLayout.jsx
   ↓
c&cLayout.jsx renders process tabs (Auto Crimping, Manual Crimping, Resistance Welding, etc.)
   ↓
AutoCrimpingView.jsx displays grid of 31 machines (C&C 01 to C&C 32, excluding 13) with live output badges
```

### Workflow C: Operator & Quality Inspector 2-Factor Authentication Gate

Prevents unauthorized machine modifications by enforcing dual sign-in before checksheet access.

```
Operator clicks a Machine Card (e.g. C&C 01)
   ↓
c&cLayout.jsx calls requestAuth(machine) from AuthContext.jsx
   ↓
If session exists: Proceeds directly to machine checksheet menu
If no session: Opens modal requesting:
  • Operator Card Number (e.g. OP-84210)
  • Quality Inspector Card Number (e.g. QI-39145)
   ↓
On submit: Saves to sessionStorage ('fme_operator_session')
   ↓
Navigates to /machine-checksheet/Bawal/CNC/C&C 01 (MachineDetailPage.jsx)
```

### Workflow D: Master Checksheet Resolution & Execution

Dynamically loads the appropriate industrial inspection form based on machine model and checksheet type.

```
User clicks a Checksheet Card (e.g. Machine 5S, Daily Checksheet, Instrument, Pokayoke, PM)
   ↓
MachineDetailPage.jsx queries getMachineModel() in src/utils/machineModels.js
   ↓
Navigates to /machine-checksheet/Bawal/CNC/<Machine>/<ChecksheetName>
   ↓
ChecksheetDetailPage.jsx resolves exact component:
  • Alpha 355 → Alpha355<Form>
  • Kodera TRD 301 / TRD 411F → Trd301<Form> / Trd411f<Form>
  • ShinMaywa G252 / G253 / G448 → G252<Form> / G253<Form> / G448<Form>
  • JAM → Jam<Form>
  • Manual Crimping → Mcr<Form>
  • Resistance Welding → ResistanceWelding<Form>
   ↓
User edits values / uploads photos → Data persisted to localStorage → Printable via window.print()
```

### Workflow E: Preventive Maintenance (PM) & Observation Action Plan Switcher

Seamlessly toggles between physical machine inspection checks and the corrective/permanent action table.

```
Technician opens 'Machine PM checksheet'
   ↓
ChecksheetDetailPage mounts AutoCrimpingPmContainer.jsx
   ↓
Initial view: Model-specific PM inspection checksheet (lubrication, mechanical wear, air pressures)
   ↓
Technician spots an abnormality → Clicks 'Observation Checksheet' tab on PmObservationSwitcher
   ↓
Container switches view to ObservationChecksheet.jsx
   ↓
Technician logs abnormality, corrective action, responsible person, and target date
   ↓
Clicks 'Save' → Serialized into localStorage under key 'fme_observation_pm_<machine>'
```

## 3. File & Folder Map: Why Every File Exists & What It Does

### Core Entry & Configuration

| File Path | Why It Exists & What It Does |
|---|---|
| `index.html` | Root HTML entry point. Configures page titles, favicons, Google Fonts, and mounts the `#root` container. |
| `src/main.jsx` | Application bootstrapper. Initializes React 19 root and renders `<App />` in StrictMode. |
| `src/App.jsx` | Central router. Defines all application paths, nested checksheet routes, shorthand redirects, and binds AuthProvider. |
| `src/MainLayout.jsx` | Main frame layout. Manages the responsive sidebar and navbar; hides both on checksheet pages for full-screen data entry. |
| `src/index.css` | Global styling foundation. Houses Tailwind directives, FME brand colors, scrollbar overrides, and print styles. |
| `src/utils/machineModels.js` | Model dictionary. Maps numerical machine IDs (e.g. C&C 01, 14, 25) to exact manufacturer models (Alpha_355, TRD_411F, etc.). |
| `package.json & vite.config.js` | Project dependencies, build scripts (`dev`, `build`, `lint`), and Vite React compilation settings. |
| `tailwind.config.js & postcss.config.js` | Tailwind utility scanner configuration and PostCSS Autoprefixer plugin registration. |
| `.oxlintrc.json` | Linter rules. Enforces React Hook safety and export hygiene using Oxlint. |

### Global Components & State Management

| File Path | Why It Exists & What It Does |
|---|---|
| `src/context/AuthContext.jsx` | Two-factor shop-floor authentication. Collects and stores Operator Card No. and Quality Inspector Card No. in `sessionStorage`. |
| `src/components/Navbar.jsx` | Top header bar. Dynamically generates breadcrumb trails from URL segments and displays active operator credentials. |
| `src/components/Sidebar.jsx` | Left-hand navigation drawer. Provides collapsible route links to Dashboard, Machine Checksheet, and global session logout. |
| `src/dashboard/OperationsDashboard.jsx` | Main telemetry engine. Renders cascading dropdown filters, duration selectors, and 3 responsive SVG output and runtime charts. |
| `src/dashboard/DashboardGraphs.jsx` | Alternative/legacy charts prototype. Contains shift-by-shift hourly plan targets and OEE curve demonstrations. |

### Checksheet Navigation & Shared Engines

| File Path | Why It Exists & What It Does |
|---|---|
| `src/machine-checksheets/UnitSelectionPage.jsx` | Plant selection screen. Allows switching between Bawal Unit (primary) and Gujrat Unit (expansion). |
| `src/machine-checksheets/OperationalUnitSelectionPage.jsx` | Department selector. Cards for CNC Machining (31 machines), Assembly Lines, and SRC Quality Hub. |
| `src/machine-checksheets/OperationalUnitRouter.jsx` | Department dispatcher. Inspects the URL parameter and renders the respective department layout. |

| | |
|---|---|
| `src/machine-checksheets/MachineDetailPage.jsx` | Machine checksheet menu page. Determines model from URL and displays available inspection cards. |
| `src/machine-checksheets/ChecksheetDetailPage.jsx` | Master checksheet resolver. Evaluates model and form name to load the exact checksheet component. |
| `src/machine-checksheets/shared/ChecksheetHeaderNav.jsx` | Standard header bar for all checksheets. Contains Back, Print, Reset, and Document Reference titles. |
| `src/machine-checksheets/shared/AutoCrimpingPmContainer.jsx` | PM container wrapper. Manages the active view state between model PM checksheets and the observation sheet. |
| `src/machine-checksheets/shared/PmObservationSwitcher.jsx` | Toggle button pill. Allows one-click switching between PM Checksheet and Observation Action Plan. |
| `src/machine-checksheets/shared/ObservationChecksheet.jsx` | Corrective & permanent action plan. Records PM abnormalities, responsible parties, and target dates in `localStorage`. |
| `src/machine-checksheets/shared/MicrometerComponentGuide.jsx` | Interactive micrometer guide. Visual callouts for plunger/blade micrometers with custom image upload and compression. |
| `src/machine-checksheets/shared/FiveSPhotoCell.jsx & FiveSHowCell.jsx` | Reusable 5S cells. `FiveSHowCell` renders cleaning tool icons; `FiveSPhotoCell` handles standard verification photos. |
| `src/machine-checksheets/shared/BreakdownSummary.jsx` | Form FRM-WH-QA-241. Comprehensive incident log for machine breakdowns, downtime, 5M changes, and parts replaced. |
| `src/machine-checksheets/shared/CcProcessProductionSummary.jsx` | Production summary form. Logs hourly targets versus actual output and tracks 14 standardized downtime categories. |
| `src/machine-checksheets/shared/OperatorMachineSummary.jsx` | Telemetry summary card. Displays active shift runtimes, breakdown downtime, and total shift production. |
| `src/machine-checksheets/shared/ChecksheetNotAdded.jsx` | Fallback component. Displays an informative empty-state screen when a checksheet is not yet configured. |

### Cutting & Crimping (CNC) Process Views & Model Forms

| File Path / Folder | Why It Exists & What It Does |
|---|---|
| `src/machine-checksheets/units/c&c/c&cLayout.jsx` | CNC master view. Houses process filter tabs and displays the active process machine grid. |
| `src/machine-checksheets/units/c&c/processFilters.jsx` | Filter strip. Horizontal pills for switching between 7 manufacturing processes. |
| `src/machine-checksheets/units/c&c/GujratEmptyCncPage.jsx` | Notification page. Displays "No Machines Added to Gujrat Unit" when accessing CNC under Gujrat. |
| `src/machine-checksheets/units/c&c/autoCrimping/AutoCrimpingView.jsx` | Primary shop-floor grid. Renders 31 C&C machine cards with live throughput rates, models, and health states. |
| `src/machine-checksheets/units/c&c/autoCrimping/<model>/`<br>`alpha355, trd301, trd411f, g252, g253, g448, jam` | Model-specific inspection suites. Each folder contains 6 dedicated forms customized for that machine model:<br>• `DailyChecksheet.jsx`: Daily operational checks & abnormality log.<br>• `FiveSChecksheet.jsx`: 5S cleanliness audit with visual method icons.<br>• `InspectionReportSheet.jsx`: Crimp heights, widths, and pull-test verification.<br>• `InstrumentChecksheet.jsx`: Calibration verification for micrometers and calipers.<br>• `MachinePmChecksheet.jsx`: Monthly preventive maintenance points.<br>• `PokayokeChecksheet.jsx`: Poka-Yoke sensors, cameras, and CFM testing. |
| `src/machine-checksheets/units/c&c/manualCrimping/` | Manual Crimping overview (`ManualCrimpingView.jsx`) and MCR/Mecal-05 checksheet suite (`mcr/`) including predictive maintenance. |
| `src/machine-checksheets/units/c&c/resistanceWelding/` | Resistance Welding overview (`ResistanceWeldingView.jsx`) and 7 R/SW checksheets including visual PM with 14 electrode inspection guides. |

| | |
|---|---|
| `src/machine-checksheets/units/c&c/twisting/, blueTaping/, headSinking/, jointCrimping/` | Secondary process views. Informative placeholder cards notifying operators that digital checksheets are currently pending setup. |
| `src/machine-checksheets/units/assembly/ & src/` | Department layouts. `assemblyLayout.jsx` and `srcLayout.jsx` provide section landing pages (files in `allChecksheets/` are upcoming placeholders). |

## 4. Data Storage Architecture & Developer Run Guide

### Storage Strategy (Pure Client-Side Persistence)

- **User Authentication:** Managed via `sessionStorage` (key: `fme_operator_session`). Automatically logs out the operator when the browser tab is closed for workstation security.
- **Checksheet Records & Inspection Data:** Automatically persisted in `localStorage` (keys: `mcr_*`, `rw_*`, `fme_observation_pm_*`). Ensures entered measurements and notes survive page refreshes.
- **Custom Verification Photos:** Compressed via HTML Canvas (1200px max, JPEG 0.85) to avoid browser storage quotas, stored as Base64 strings in `localStorage`.

### Standard Developer Commands

| Command | Purpose |
|---|---|
| `npm install` | Installs all Node dependencies (React 19, Vite, Tailwind CSS, Lucide React, Oxlint). |
| `npm run dev` | Starts the local development server at `http://localhost:5173/` with live hot reloading. |
| `npm run lint` | Runs Oxlint to check JavaScript and React Hook rules across all source files. |
| `npm run build` | Compiles, bundles, and minifies the application into production-ready assets inside `dist/`. |
| `npm run preview` | Spins up a lightweight local server to test and preview the production `dist/` build. |